

★ WEEK 11 · PART 1 of 3

Continuous-Time Models: Brownian Motion and SDEs

When Time Flows Continuously

Brownian motion · the increment dW · SDEs · GBM · Ito's lemma

Every model so far ticked in discrete steps. Now we let time flow continuously — the language of modern finance. It begins with Brownian motion and the stochastic differential equations built on it.

1

PART 1 · BROWNIAN MOTION & SDEs

The Language of Continuous Time

Why continuous time, Brownian motion and its increment dW , stochastic differential equations, geometric Brownian motion for prices, and Ito's lemma — the stochastic chain rule.

The Thesis 60 Years Ahead of Its Time — Part 1

A student models the market with random motion

In 1900, Louis Bachelier's PhD thesis modeled stock-price fluctuations as what we now call Brownian motion — five years before Einstein's famous physics paper on the same phenomenon.

What he did

- Treated price changes as accumulating random increments.
- Derived the mathematics of Brownian motion for markets.
- Anticipated much of modern option pricing.

The reception

- It was far ahead of its time and largely overlooked.
- The tools to appreciate it did not yet exist.
- It sat mostly unread for half a century.

The Thesis 60 Years Ahead of Its Time — Part 2

Rediscovery and the continuous-time revolution

Rediscovered mid-century — notably by Paul Samuelson — Bachelier's idea, refined into geometric Brownian motion, underpinned the Black-Scholes-Merton revolution and modern continuous-time finance.

The revival

- Economists rediscovered the thesis in the 1950s.
- Brownian motion became the base for asset-price models.
- It powered the option-pricing breakthroughs of the 1970s.

The lesson for us

- Continuous time is the native language of modern finance.
- Its foundation is a single object: Brownian motion.
- From it we build SDEs — and a clean mean-reversion model.

From Discrete Steps to Continuous Flow

Where Weeks 1-10 leave us

1

Everything so far ticked discretely

ARIMA, GARCH, VAR, Kalman, HMMs — all evolve one step at a time, on a fixed grid of observations.

2

But markets don't wait for the bell

Prices move continuously; derivatives are priced in continuous time. We need models that flow rather than step.

3

Continuous time needs new tools

Brownian motion, stochastic differential equations, and Ito's calculus — the machinery of continuous-time finance, and the home of the OU mean-reversion model.

Part 1 in one line: build the continuous-time language — Brownian motion, SDEs, and Ito.

Why Move to Continuous Time?

Markets don't really wait for the closing bell

Discrete models pretend the world updates on a fixed clock. But trades happen at every instant, and the whole theory of derivatives lives in continuous time. Letting time flow gives cleaner, more powerful models.

What discrete time misses

- Prices move between the grid points, not just on them.
- Option pricing and hedging are continuous-time ideas.
- The math is often cleaner in the continuous limit.

What continuous time buys

- Elegant models with closed-form results.
- A natural home for calculus-based reasoning.
- The framework the rest of quantitative finance uses.

Brownian Motion (the Wiener Process)

The source of continuous-time randomness

$$W_0 = 0, \quad W_t - W_s \sim \mathcal{N}(0, t - s), \quad \text{independent increments, continuous paths}$$

Its properties

- It starts at zero and moves continuously.
- Increments over disjoint intervals are independent.
- An increment over length h is Gaussian with variance h .

Why it is fundamental

- It is the continuous-time limit of a random walk.
- It is the raw randomness every SDE is built from.
- Its paths are continuous but nowhere smooth.

The Increment dW

The atom of stochastic calculus

$$dW_t \sim \mathcal{N}(0, dt) \quad \Rightarrow \quad (dW_t)^2 = dt \quad (\text{the key heuristic})$$

What dW is

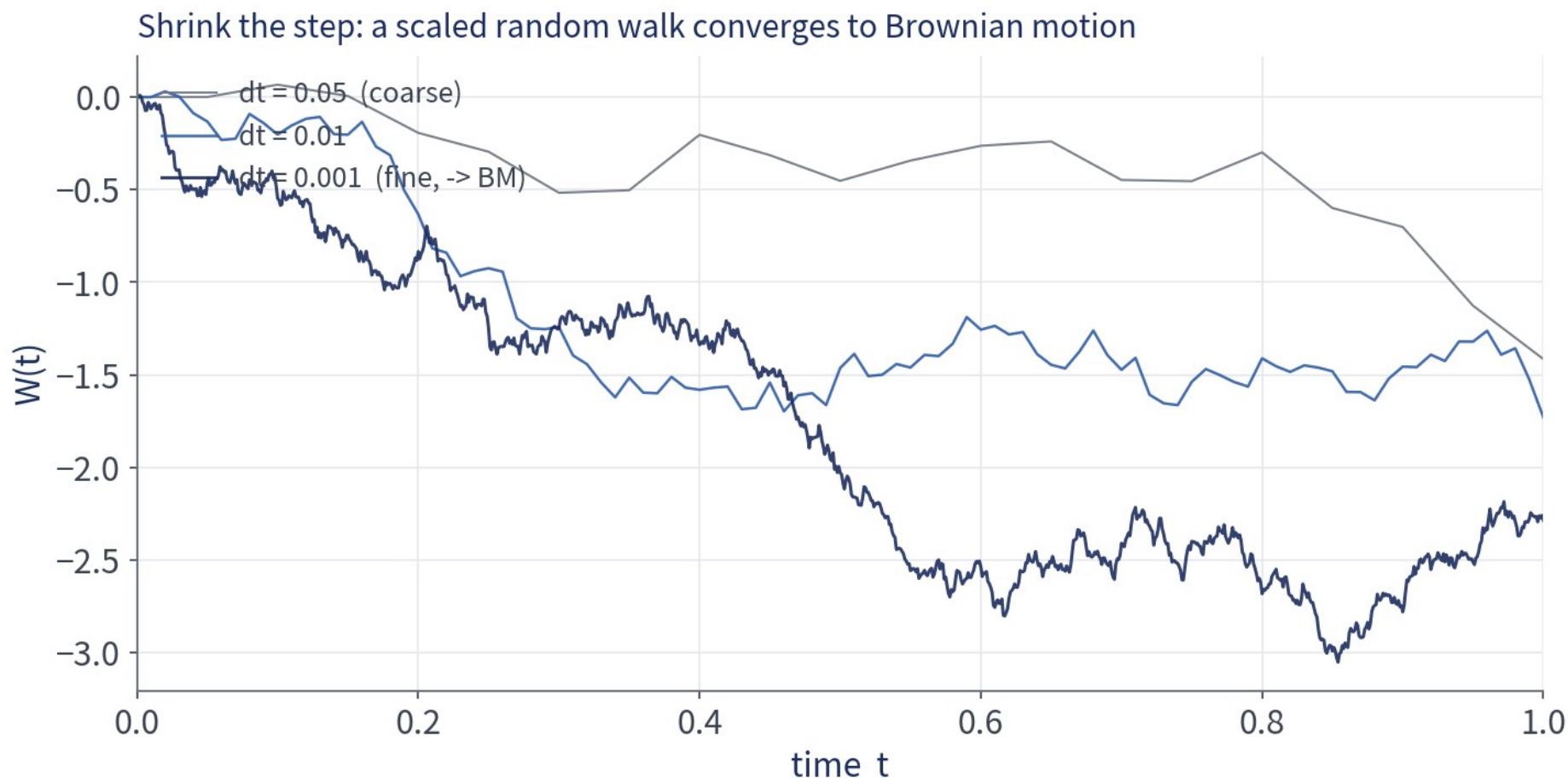
- An infinitesimal Brownian increment over dt .
- It is Gaussian with mean zero and variance dt .
- Its standard deviation scales as the square root of dt .

The key heuristic

- Squaring it gives dt , not a higher-order term.
- So (dW) squared cannot be discarded like (dt) squared.
- This single fact is what makes stochastic calculus different.

Random Walk Becomes Brownian Motion

Shrink the time step and a suitably scaled random walk fills in to a continuous Brownian path. Brownian motion is exactly this limit — the continuous-time cousin of the random walk.



Key Properties of Brownian Motion

Strange, but essential

What is true

- Continuous paths — no jumps.
- Independent, stationary Gaussian increments.
- Variance grows linearly with time.

What is surprising

- The paths are nowhere differentiable — infinitely jagged.
- So you cannot take an ordinary derivative dW/dt .
- That is why we work with increments dW , not slopes.

Stochastic Differential Equations

Trend plus randomness, in one line

$$dX_t = \underbrace{a(X_t, t) dt}_{\text{drift}} + \underbrace{b(X_t, t) dW_t}_{\text{diffusion}}$$

The two terms

- The drift a is the deterministic trend per unit time.
- The diffusion b scales the random Brownian shock.
- Both can depend on the current state and time.

Why this is general

- Almost every continuous-time model has this form.
- Choose a and b , and you choose the model.
- It is the continuous-time analogue of a recursion.

Reading an SDE

Drift steers, diffusion shakes

The drift term

- It is where the process is heading on average.
- Set it to zero and there is no trend.
- It captures growth, decay, or pull toward a level.

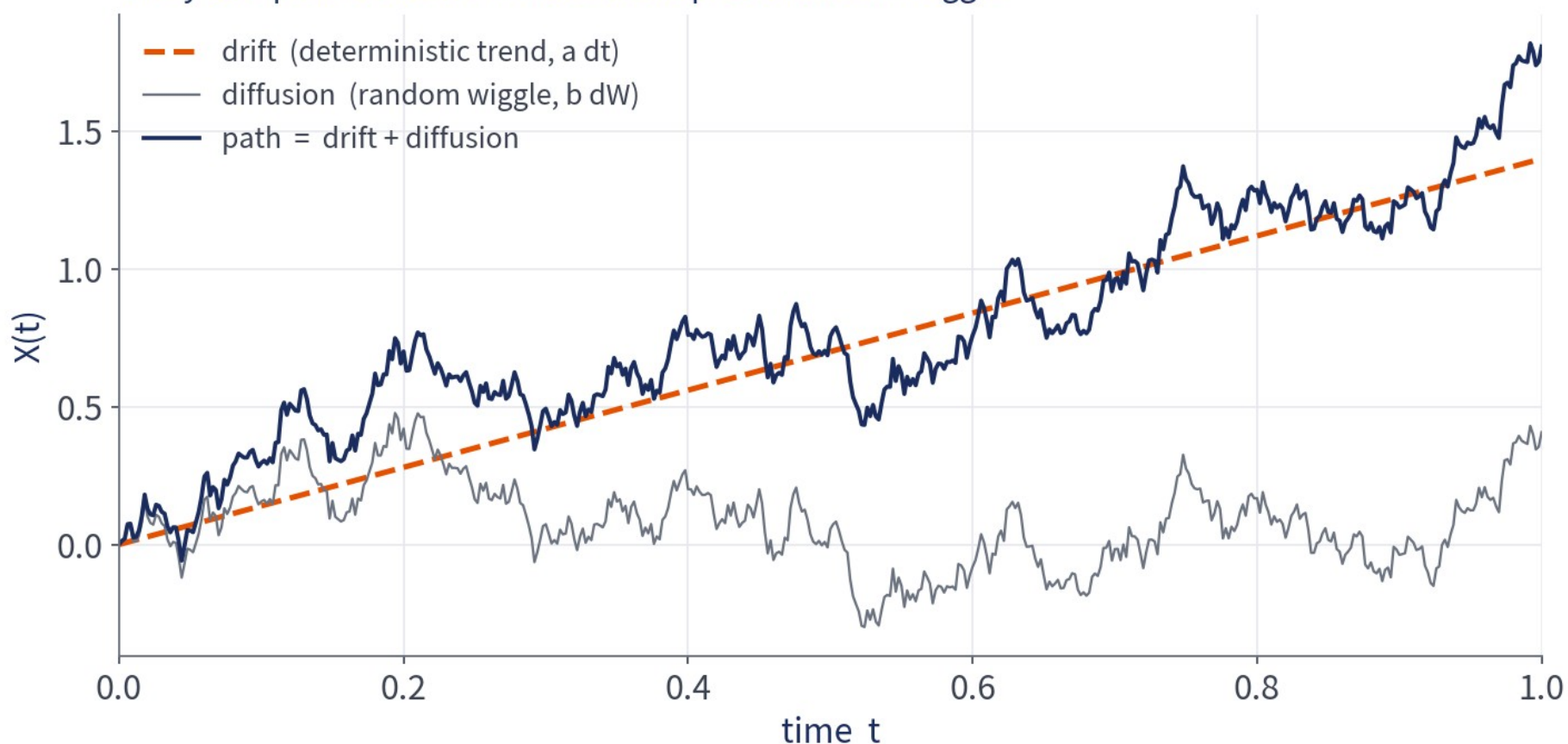
The diffusion term

- It is the size of the random fluctuations.
- Larger diffusion means a wilder path.
- It multiplies the Brownian increment dW .

Drift Plus Diffusion

Every SDE path splits into a smooth deterministic trend (drift) and a random wiggle around it (diffusion). Their sum is the realized path — trend plus noise, in continuous time.

Every SDE path is a deterministic trend plus a random wiggle



Geometric Brownian Motion

The standard model for an asset price

$$\text{GBM: } dS_t = \mu S_t dt + \sigma S_t dW_t$$

The specification

- Both drift and diffusion are proportional to the price.
- So returns, not prices, have constant statistics.
- The price stays positive and grows multiplicatively.

Why it fits prices

- Percentage moves, not dollar moves, are what's stable.
- It gives lognormally-distributed future prices.
- It is the engine behind the Black-Scholes formula.

Why GBM for Prices

Positive, proportional, tractable

Its virtues

- The price can never go negative.
- Moves scale with the level — a 1% move is a 1% move.
- It has clean closed-form solutions.

The intuition

- Model the log-price as Brownian motion with drift.
- Exponentiate to get a positive price.
- Simple, and good enough as a first approximation.

GBM's Strengths and Its Blind Spots

A great default, not a universal truth

What it gets right

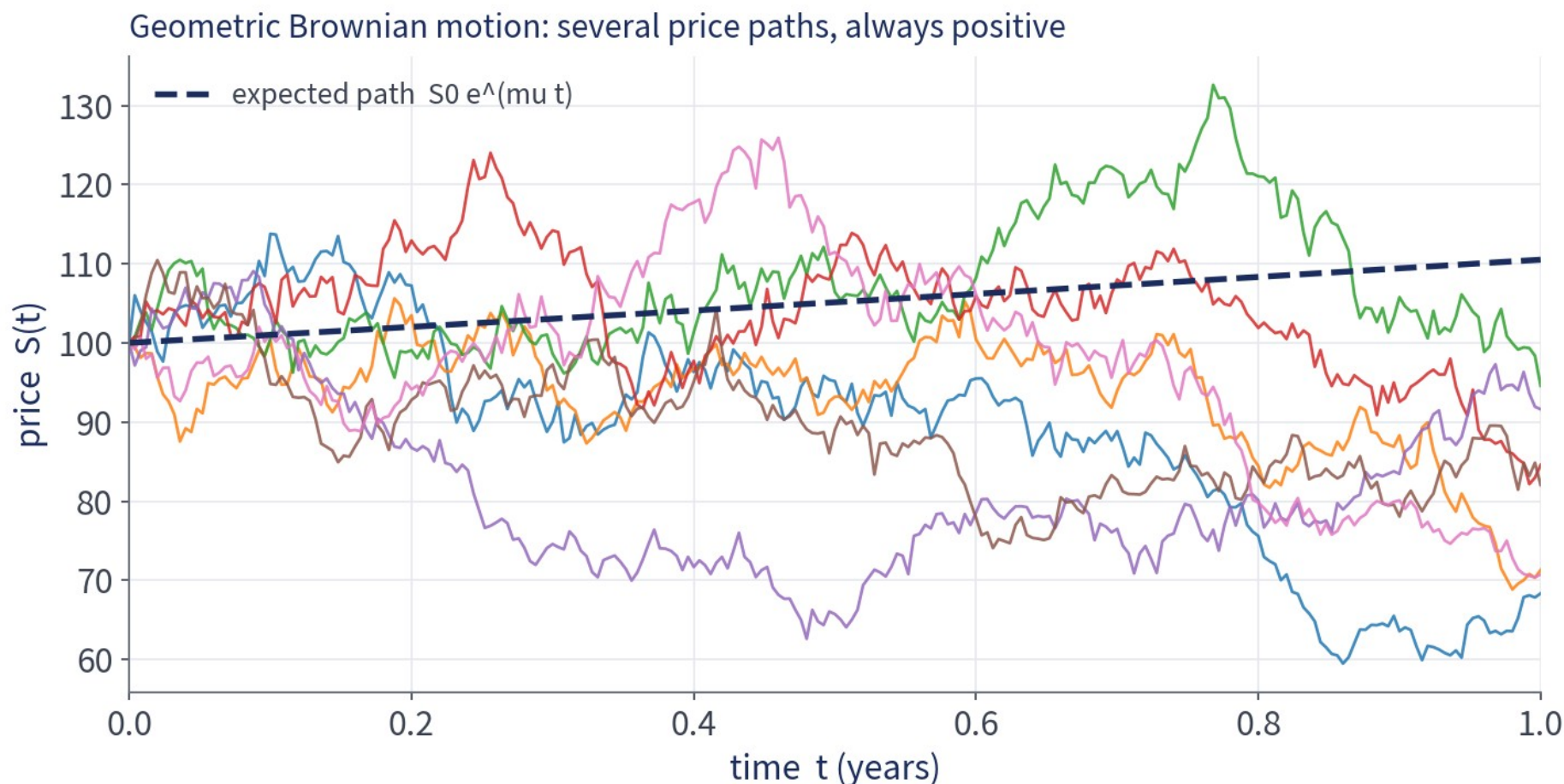
- Positivity and proportional moves.
- A tractable, well-understood benchmark.
- A sensible baseline for many assets.

What it misses

- Constant volatility — no clustering (recall GARCH).
- Thin (Gaussian) tails — no crashes.
- No mean reversion — prices wander without an anchor.

Geometric Brownian Motion in Action

Several GBM price paths from the same start: always positive, fanning out into a lognormal spread of outcomes around the expected exponential-growth path.



Ito's Lemma: The Stochastic Chain Rule

Differentiating a function of a random process

To work with functions of a random process — like the log of a price — you need a chain rule that accounts for the roughness of Brownian paths. That rule is Ito's lemma, and it carries an extra term.

The problem

- Ordinary calculus assumes smooth paths.
- Brownian paths are nowhere differentiable.
- So the usual chain rule does not apply.

The fix

- Ito's lemma extends the chain rule to SDEs.
- It adds a second-order term from the $(dW)^2 = dt$ fact.
- It is the workhorse of continuous-time finance.

The Extra Term

Where Ito departs from ordinary calculus

$$\text{Ito: } df = \left(f_t + a f_x + \frac{1}{2}b^2 f_{xx} \right) dt + b f_x dW_t \quad \leftarrow \quad \frac{1}{2}b^2 f_{xx} \text{ is the extra term}$$

The new piece

- Beyond the usual terms sits a second-derivative term.
- It is proportional to the diffusion squared.
- It comes directly from (dW) squared equalling dt .

Why it matters

- Ignore it and your dynamics are simply wrong.
- It is what makes log-GBM have a corrected drift.
- Every continuous-time derivation leans on it.

Why Ito Differs From Ordinary Calculus

Roughness leaves a footprint

Ordinary calculus

- Smooth paths: squared increments vanish faster than dt .
- So only first-order terms survive.
- The familiar chain rule holds.

Ito calculus

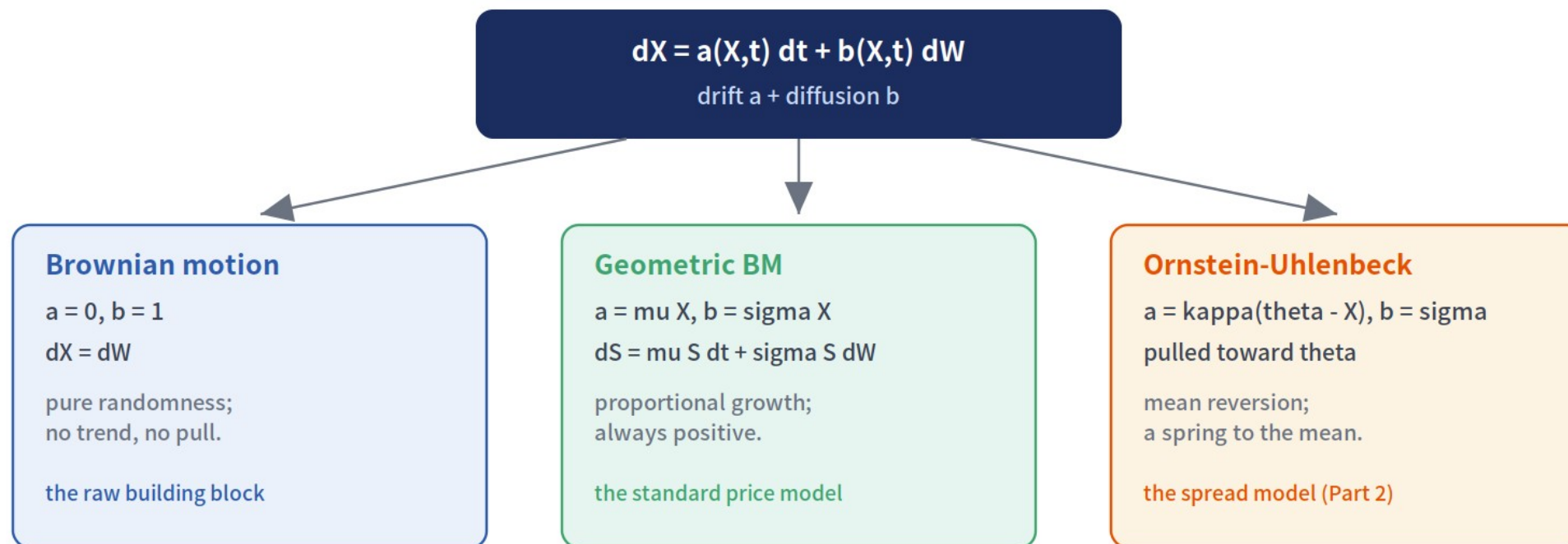
- Brownian squared increments are of order dt , not smaller.
- So a second-order term survives into the differential.
- That surviving term is the whole difference.

One Recipe, a Family of Models

Fix the drift and diffusion and the same SDE template becomes Brownian motion, geometric Brownian motion, or the Ornstein-Uhlenbeck process — the mean-reverting model we build next.

One Recipe, a Family of Models

Choose the drift and diffusion, and the same SDE template becomes a different model.



What Parts 2 and 3 Add

From randomness to mean reversion

Part 2 — the OU process

- An SDE with a drift that PULLS toward a mean.
- Continuous-time mean reversion, with three parameters.
- It is the continuous cousin of an AR(1).

Part 3 — in practice

- Model the pairs spread as an OU process (Stage V).
- Estimate its parameters and half-life.
- The cautions, and the bridge to interest-rate models.

Simulate Brownian Motion and GBM

Direct Claude Code to build the basics from scratch

GOAL Simulate Brownian motion and GBM, and see the random-walk-to-Brownian limit with your own eyes.

[INPUT] Prompt · Claude Code

Write and run `sde_basics.py` that:

1. simulates Brownian motion by cumulating Gaussian increments with variance dt ;
2. shows that shrinking dt converges to a smooth Brownian path (overlay a few step sizes);
3. simulates GBM $dS = \mu S dt + \sigma S dW$ via the exact log-Euler scheme, plotting paths;
4. checks that the log-returns are Gaussian with the right mean and variance.

Confirm GBM stays positive and returns look normal.

How to work with it

- Brownian variance should grow linearly with time.
- Smaller dt gives a finer, continuous-looking path.
- GBM paths must stay strictly positive.
- Keep it; Part 2 swaps in mean reversion.

Deliverable: `sde_basics.py` + Brownian and GBM paths, and a normality check on returns.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

What is the variance of a Brownian increment over a time step dt ?

dt

Q2

In an SDE, what do the drift and diffusion terms each represent?

**Trend and
randomness**

Q3

Why does Ito's lemma carry an extra second-order term?

Because $(dW)^2 = dt$

All three come from the increment dW , the SDE recipe, and Ito's key fact.

Three Misconceptions to Avoid

Where continuous-time intuition slips

Common myths

- "Brownian paths have a well-defined velocity."
- "You can differentiate a function of a process as usual."
- "GBM captures crashes and volatility clustering."

The reality

- Brownian paths are nowhere differentiable — no velocity.
- Use Ito's lemma; it adds a second-order term.
- GBM has constant volatility and thin tails — no crashes.

Three Things to Carry Forward

Part 1 in one screen

1

Brownian motion is the building block

The continuous-time limit of a random walk: continuous, nowhere smooth, with Gaussian increments whose variance is dt . Its increment dW is the atom of stochastic calculus.

2

An SDE is drift plus diffusion

Choose a deterministic trend and a random diffusion and you specify a model. GBM — proportional drift and diffusion — is the standard, positive, lognormal price model.

3

Ito's lemma is the stochastic chain rule

Because $(dW)^2$ equals dt , differentiating a function of a process carries an extra second-order term. It is the workhorse of every continuous-time derivation.

Next — Part 2: the Ornstein-Uhlenbeck process, continuous-time mean reversion for a spread.